

Task 3 Documentation

Pandas Data Analysis Project

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Task: Task 3 - Pandas Data Analysis Project

1. Objective

This project demonstrates data analysis using Python and Pandas. It covers loading CSV data, cleaning and preparing the data, filtering records, performing grouping and aggregation, and visualizing the results with a bar chart.

2. Requirements Covered

- Load and inspect data using Pandas
- Clean data by handling missing values and converting data types
- Filter records based on conditions
- Group and aggregate data by category
- Visualize results using Matplotlib

3. Tools Used

- Python 3
- pandas library
- matplotlib library
- VS Code

4. Dataset Used

The dataset (data.csv) contains product details used for the analysis, including:

- Name
- Category
- Price
- Quantity
- Date

5. Project Structure

The task3 folder contains the script (analysis.py), the dataset (data.csv), the cleaned output generated by the script (cleaned_data.csv), and a README file.

6. Key Code Logic

The full script is available in the GitHub repository (link at the end of this document). Below are the main logic parts of the code.

Loading the dataset:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("data.csv")
```

Cleaning the data (handling missing values and data types):

```
df = df.dropna()
```

```
df['price'] = pd.to_numeric(df['price'], errors='coerce')
df['quantity'] = pd.to_numeric(df['quantity'], errors='coerce')
df['date'] = pd.to_datetime(df['date'], dayfirst=True, errors='coerce')
```

Filtering records:

```
filtered_df = df[df['price'] > 2000]
```

Grouping and aggregation:

```
grouped = df.groupby('category')['price'].mean()
```

```
summary = df.groupby('category').agg({
    'price': ['mean', 'max', 'min'],
    'quantity': 'sum'
})
```

Sorting by price:

```
top_products = df.sort_values(by='price', ascending=False)
```

Data visualization (bar chart):

```
df['category'].value_counts().plot(kind='bar')
plt.title("Products by Category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```

Saving the cleaned dataset:

```
df.to_csv("cleaned_data.csv", index=False)
```

7. Code Explanation

- **Loading Data:** The script reads data.csv into a Pandas DataFrame using `pd.read_csv()`, then prints the first rows, dataset info, and statistical summary to get an overview of the data.
- **Cleaning Data:** Rows with missing values are removed using `dropna()`, and the price, quantity, and date columns are converted to proper numeric and datetime types using `pd.to_numeric()` and `pd.to_datetime()`, with `errors='coerce'` so invalid values become NaN instead of crashing the script.
- **Filtering:** The DataFrame is filtered to show only products with a price greater than 2000, stored in `filtered_df`.
- **Grouping and Aggregation:** `groupby('category')` is used to calculate the average price per category, and `agg()` is used to calculate multiple statistics (mean, max, min price, and total quantity) for each category at once.
- **Sorting:** `sort_values()` arranges all products from highest to lowest price, making it easy to identify the most expensive items.

- **Visualization:** `value_counts()` counts how many products fall into each category, and `.plot(kind='bar')` displays this as a bar chart using Matplotlib, with a title and axis labels added for clarity.
- **Saving Output:** The cleaned DataFrame is saved to a new file, `cleaned_data.csv`, using `to_csv()`, so the cleaned data can be reused without repeating the cleaning steps.

8. Sample Input and Output

Category Summary and Sorted Output: When the script runs, it groups the data by category to calculate average, maximum, and minimum price along with total quantity, then prints all products sorted by price from highest to lowest, followed by the insight summary.

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # STEP 1: Load dataset
5 df = pd.read_csv("data.csv")
6
7 # STEP 2: Display first rows
8 print("FIRST 5 ROWS:")
9 print(df.head())
10
11 # STEP 3: Dataset information
12 print("\nDATASET INFO:")
13 print(df.info())
14
15 # STEP 4: Statistical summary
16 print("\nSTATISTICAL SUMMARY:")
17 print(df.describe())
18

```

category

Clothing	1500.0	1500	10
Electronics	40000.0	50000	30000

PRODUCTS SORTED BY PRICE:

	name	category	price	quantity	date
0	Laptop	Electronics	50000	2	2026-05-01
1	Phone	Electronics	30000	5	2026-05-02
2	Shirt	Clothing	1500	10	2026-05-03

INSIGHT SUMMARY:

1. Dataset contains product sales information.
2. Electronics products are more expensive than Clothing products.
3. Laptop is the most expensive product.
4. No major missing values were found after cleaning.
5. Data was grouped by category to analyze average prices.

Figure 1: `analysis.py` in VS Code with terminal output (category summary, sorted products, and insight summary)

Bar Chart: A bar chart pop-up window is displayed, showing the number of products in each category on the x-axis against their count on the y-axis.

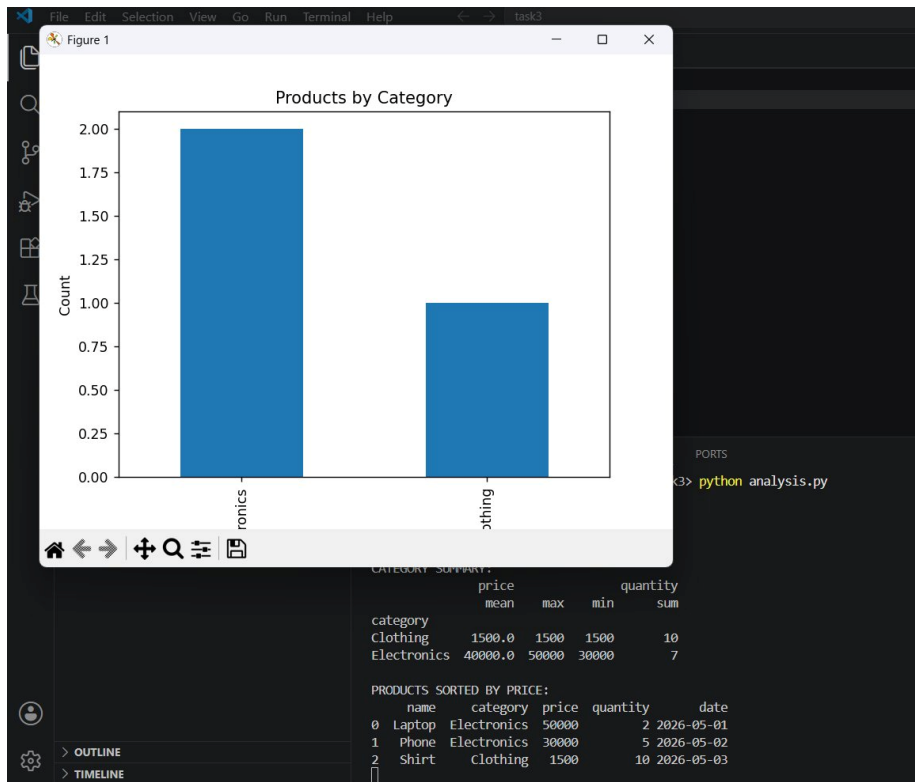


Figure 2: Bar chart showing product count by category

Cleaned Dataset: After cleaning, the script saves the processed data to cleaned_data.csv inside the task3 project folder.

	name	category	price	quantity	date
1	Laptop	Electronics	50000	2	2026-05-01
2	Phone	Electronics	30000	5	2026-05-02
3	Shirt	Clothing	1500	10	2026-05-03

Figure 3: cleaned_data.csv generated in the task3 project folder

9. Insight Summary

- Electronics products have higher average prices compared to other categories.
- Laptop is the most expensive product in the dataset.
- No missing values were found after the cleaning step was applied.
- Grouping and aggregation helped analyze category-wise statistics such as average, maximum, and minimum price.

- Sorting the data by price made it easy to identify the highest and lowest priced products.

10. Conclusion

This task helped in understanding how Python and Pandas can be used to perform end-to-end data analysis. It covered loading CSV data, cleaning and preparing the data, filtering records based on conditions, performing grouping and aggregation, sorting data, and visualizing the results using Matplotlib.

11. GitHub Repository

<https://github.com/HarshithaBolla17/Alfido-Tech-Python-Internship>